ELEC2134 Circuits & Signals: Cheat Sheet

Maths reference

Integrals

$f(x)$	$\int f(x)\,dx$
$\sin(ax)$	$-\frac{1}{a}\cos(ax)+C$
$\cos(ax)$	$\frac{1}{a}\sin(ax)+C$
e^{ax}	$\frac{1}{a}e^{ax}+C$
xe^{ax}	$\frac{e^{ax}}{a^2}(ax-1)+C$
x^2e^{ax}	$\frac{e^{ax}}{a^3}(a^2x^2-2ax+2)+C$
$x\cos(ax)$	$\frac{\cos(ax)}{a^2}+\frac{x\sin(ax)}{a}+C$
$x\sin(ax)$	$\frac{\sin(ax)}{a^2}-\frac{x\cos(ax)}{a}+C$
$e^{ax}\sin(bx)$	$\frac{e^{ax}(a\sin(bx)-b\cos(bx))}{a^2+b^2}+C$
$e^{ax}\cos(bx)$	$\frac{e^{ax}(a\cos(bx)+b\sin(bx))}{a^2+b^2}+C$
$\frac{1}{x^2+a^2}$	$\frac{1}{a}\arctan\left(\frac{x}{a}\right)+C$

Rule	Formula
By parts	$\int u\,dv = uv - \int v\,du$
One period ($\omega_0 = 2\pi/T$, integer $n \neq 0$)	$\int_0^T \sin(n\omega_0 t)\,dt = \int_0^T \cos(n\omega_0 t)\,dt = 0$
One period, $t\sin$	$\int_0^T t\sin(n\omega_0 t)\,dt = -\frac{T}{n\omega_0}$
One period, $t\cos$	$\int_0^T t\cos(n\omega_0 t)\,dt = 0$
One period, $x^2\cos$	$\int_0^T x^2\cos(nx)\,dx = \left[\frac{x^2}{n}\sin(nx)+\frac{2x}{n^2}\cos(nx)-\frac{2}{n^3}\sin(nx)\right]_0^T$
One period, $x^2\sin$	$\int_0^T x^2\sin(nx)\,dx = \left[-\frac{x^2}{n}\cos(nx)+\frac{2x}{n^2}\sin(nx)+\frac{2}{n^3}\cos(nx)\right]_0^T$
Orthogonality, $\cos\cdot\sin$	$\int_0^T \cos(m\omega_0 t)\sin(n\omega_0 t)\,dt = 0 \quad \text{for all } m,n$
Orthogonality, $\sin\cdot\sin$ and $\cos\cdot\cos$	$\int_0^T \sin(m\omega_0 t)\sin(n\omega_0 t)\,dt = \int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)\,dt = \begin{cases} 0 & m\neq n \\ T/2 & m=n \end{cases}$
Gamma-type	$\int_0^\infty t^n e^{-at}\,dt = \frac{n!}{a^{n+1}}$

Trig identities

Identity	Formula
Parity	$\sin(-\alpha) = -\sin\alpha \qquad \cos(-\alpha) = \cos\alpha$
Shift by $\pi/2$	$\sin(\alpha+\pi/2) = \cos\alpha \qquad \cos(\alpha+\pi/2) = -\sin\alpha$
Shift by π	$\sin(\alpha\pm\pi) = -\sin\alpha \qquad \cos(\alpha\pm\pi) = -\cos\alpha$
Pythagorean, double sin	$\sin^2\alpha+\cos^2\alpha = 1 \qquad \sin 2\alpha = 2\sin\alpha\cos\alpha$
Sum / difference, sin	$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
Sum / difference, cos	$\cos(\alpha\pm\beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
Product, cos cos	$\cos\alpha\cos\beta = \frac{1}{2}\cos(\alpha+\beta)+\frac{1}{2}\cos(\alpha-\beta)$
Product, sin sin	$\sin\alpha\sin\beta = \frac{1}{2}\cos(\alpha-\beta)-\frac{1}{2}\cos(\alpha+\beta)$
Product, sin cos	$\sin\alpha\cos\beta = \frac{1}{2}\sin(\alpha+\beta)+\frac{1}{2}\sin(\alpha-\beta)$
Double angle, cos	$\cos 2\alpha = 2\cos^2\alpha-1 = 1-2\sin^2\alpha = \cos^2\alpha-\sin^2\alpha$
Power reduction	$\sin^2\alpha = \frac{1}{2}(1-\cos 2\alpha) \qquad \cos^2\alpha = \frac{1}{2}(1+\cos 2\alpha)$
Combine into one cos	$A\cos\alpha + B\sin\alpha = \sqrt{A^2+B^2}\cos(\alpha-\operatorname{atan2}(B,A))$

Euler, complex numbers, decibels

Item	Formula
Euler	$e^{\pm j\alpha} = \cos\alpha \pm j\sin\alpha$
sin, cos from exponentials	$\sin\alpha = \frac{e^{j\alpha}-e^{-j\alpha}}{2j} \qquad \cos\alpha = \frac{e^{j\alpha}+e^{-j\alpha}}{2}$
Polar form	$z = a+jb = re^{j\theta} \quad r = \sqrt{a^2+b^2} \quad a = r\cos\theta \quad b = r\sin\theta$
Decibels	$\text{dB} = 20\log_{10}\left(\frac{V_{\text{out}}}{V_{\text{in}}}\right) = 10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)$
-3 dB point	$1/\sqrt{2} = 0.707$ gain
Common dB values	$10\log_{10}2 = 3$ dB; $10\log_{10}(1/2) = -3$ dB (half power); $+10$ dB per $\times 10$ in power, $+20$ dB per $\times 10$ in voltage ($20\log_{10}10 = 20$)

Phase of a complex number (atan2, quadrants)

$$\alpha = \arctan(|b|/|a|) \quad (0 \le \alpha \le \pi/2)$$

Location of $z = a+jb$	$\theta = \angle z$
$a>0, b\ge 0$ (Q1)	α
$a<0, b\ge 0$ (Q2)	$\pi-\alpha$
$a<0, b<0$ (Q3)	$-(\pi-\alpha) = \alpha-\pi$
$a>0, b<0$ (Q4)	$-\alpha$
$a=0$ (α undefined)	$\pm\frac{\pi}{2}$ for $b>0$ / $b<0$

Example. $z = -3 + j4$ (Q2): $\alpha = \arctan(4/3) = 53.1°$, so $\theta = 180° - 53.1° = 126.9°$.
$z = -3 - j4$ (Q3): same α, so $\theta = 53.1° - 180° = -126.9°$. A bare $\arctan(b/a)$ gives $-53.1°$ for the first and $+53.1°$ for the second, both wrong.

Laplace transform

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)e^{-st}\,\mathrm{d}t \qquad s = \sigma + j\omega$$

σ is the exponential damping, ω the frequency. The **ROC** is the set of σ for which the integral converges. If $f(t) = 0$ for $t < 0$ and the ROC includes the $j\omega$-axis, then $F(\omega) = F(s)|_{s=j\omega}$ (Fourier = Laplace on $\sigma = 0$).

$f(t)$	$F(s)$	$f(t)$	$F(s)$
$\delta(t)$	1	$e^{-at}\sin\omega t$	$\dfrac{\omega}{(s+a)^2+\omega^2}$
$\delta(t-\tau)$	$e^{-s\tau}$	$e^{-at}\cos\omega t$	$\dfrac{s+a}{(s+a)^2+\omega^2}$
$u(t)$	$\dfrac{1}{s}$	$\sin\omega t$	$\dfrac{\omega}{s^2+\omega^2}$
$tu(t)$	$\dfrac{1}{s^2}$	$\cos\omega t$	$\dfrac{s}{s^2+\omega^2}$
t^n	$\dfrac{n!}{s^{n+1}}$	$e^{-bt}t^n$	$\dfrac{n!}{(s+b)^{n+1}}$
e^{-at}	$\dfrac{1}{s+a}$	te^{-at}	$\dfrac{1}{(s+a)^2}$

Property	$f(t) \Longleftrightarrow F(s)$
Linearity	$af_1(t) + bf_2(t) \Longleftrightarrow aF_1(s) + bF_2(s)$
Scaling	$f(at) \Longleftrightarrow \dfrac{1}{a}F\left(\dfrac{s}{a}\right)$
Time shift	$f(t-a)u(t-a) \Longleftrightarrow e^{-as}F(s)$
Frequency shift	$e^{-at}f(t) \Longleftrightarrow F(s+a)$
First derivative	$f'(t) \Longleftrightarrow sF(s) - f(0^-)$
Second derivative	$f''(t) \Longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-)$
Integration	$\displaystyle\int_0^t f(\tau)\,\mathrm{d}\tau \Longleftrightarrow \frac{F(s)}{s}$
s-differentiation	$t^n f(t) \Longleftrightarrow (-1)^n \dfrac{\mathrm{d}^n F}{\mathrm{d}s^n}$
s-integration	$\dfrac{f(t)}{t} \Longleftrightarrow \displaystyle\int_s^{\infty} F(s')\,\mathrm{d}s'$
Convolution	$f(t) * g(t) \Longleftrightarrow F(s)G(s)$

Initial and final value ($f(\infty)$ needs a finite final value, i.e. all poles of sF in the left half-plane):

$$f(0^+) = \lim_{s\to\infty} sF(s) \qquad f(\infty) = \lim_{s\to 0} sF(s)$$

Periodic function with period T (F_1 is the transform of the first period only):

$$F(s) = \frac{F_1(s)}{1 - e^{-sT}} \qquad F_1(s) = \int_0^T f(t)e^{-st}\,\mathrm{d}t$$

Example. Ramp pulse train ($T = 2$, $f_1 = 2t$ on $(0, 1)$):

$$F_1(s) = \frac{2[1-(1+s)e^{-s}]}{s^2} \qquad F(s) = \frac{2[1-(1+s)e^{-s}]}{s^2(1-e^{-2s})}$$

Inverse Laplace: partial fractions

Distinct pole p, and repeated pole of multiplicity m (term $A_k/(s-p)^k$):

$$A = [(s-p)F(s)]_{s=p}$$

$$A_k = \frac{1}{(m-k)!}\frac{\mathrm{d}^{m-k}}{\mathrm{d}s^{m-k}}[(s-p)^m F(s)]_{s=p}$$

For a complex pair keep $(Bs + C)/((s+a)^2 + \omega^2)$ and match $e^{-at}\cos$ and $e^{-at}\sin$. If $\deg N \geq \deg D$, divide first.

Example.

$$\frac{s^2+12}{s(s+2)(s+3)} = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

$$A = \frac{12}{6} = 2 \quad B = \frac{4+12}{(-2)(1)} = -8 \quad C = \frac{9+12}{(-3)(-1)} = 7$$

Example. Solve $v'' + 6v' + 8v = 2u(t)$ with $v(0) = 1$, $v'(0) = -2$:

$$(s^2+6s+8)V(s) = s + 4 + \frac{2}{s} = \frac{s^2+4s+2}{s}$$

$$V(s) = \frac{1/4}{s} + \frac{1/2}{s+2} + \frac{1/4}{s+4}$$

$$v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t)$$

ROC, causality and stability

$H(s)$ alone does not fix $h(t)$; the ROC does. The same $1/(s+a)$ comes from two different signals:

Signal	$F(s)$	ROC
$e^{-at}u(t)$ (right-sided)	$\dfrac{1}{s+a}$	$\sigma > -a$
$-e^{-at}u(-t)$ (left-sided)	$\dfrac{1}{s+a}$	$\sigma < -a$

The ROC is a vertical strip in $\sigma = \mathrm{Re}(s)$ and contains no poles. Right-sided: ROC right of the rightmost pole. Left-sided: ROC left of the leftmost pole. Two-sided: a strip between poles.

Causal: ROC is $\sigma >$ rightmost pole. **Stable:** ROC includes the $j\omega$-axis. **Causal and stable:** all poles have $\sigma < 0$.

Finding $h(t)$ from $H(s)$ and the ROC: partial fractions $H = \sum A_k/(s-p_k)$, then for each pole

p_k left of the ROC: $A_k e^{p_k t}u(t)$ $\qquad$ p_k right of the ROC: $-A_k e^{p_k t}u(-t)$

Example. $H(s) = \frac{3s-1}{(s+3)(s-2)} = \frac{2}{s+3} + \frac{1}{s-2}$:

ROC	$h(t)$	Property
$\sigma > 2$	$2e^{-3t}u(t) + e^{2t}u(t)$	causal, unstable
$-3 < \sigma < 2$	$2e^{-3t}u(t) - e^{2t}u(-t)$	stable, non-causal
$\sigma < -3$	$-2e^{-3t}u(-t) - e^{2t}u(-t)$	anti-causal, unstable

Circuits in the s-domain

	Time	V form	I form	$Z(s)$
R	$v = Ri$	$V = RI$	$I = \dfrac{V}{R}$	R
L	$v = L\dfrac{\mathrm{d}i}{\mathrm{d}t}$	$V = sLI - Li(0)$	$I = \dfrac{V}{sL} + \dfrac{i(0)}{s}$	sL
C	$i = C\dfrac{\mathrm{d}v}{\mathrm{d}t}$	$V = \dfrac{I}{sC} + \dfrac{v(0)}{s}$	$I = sCV - Cv(0)$	$\dfrac{1}{sC}$

Admittance is $Y = 1/Z$. With zero initial conditions, $Z_R = R$, $Z_L = sL$, $Z_C = 1/(sC)$. Series RLC: $(sL + R + 1/(sC))I(s) = V(s)$.

Transfer function (zero initial conditions; a circuit has many, depending on the chosen input and output):

$$H(s) = \frac{Y(s)}{X(s)} \qquad Y(s) = H(s)X(s) \qquad h(t) = \mathcal{L}^{-1}\{H(s)\}$$

H can be a voltage gain V_o/V_i, current gain I_o/I_i, impedance V/I or admittance I/V. $h(t)$ is the unit impulse response (input $\delta(t)$, $X = 1$). **ODE $\leftrightarrow$ $H(s)$:** replace $\mathrm{d}^k/\,\mathrm{d}t^k$ by s^k (zero ICs).

Example. $h(t) = e^{-t}u(t)$, $v_i(t) = 10e^{-2t}u(t)$:

$$V_o(s) = H(s)V_i(s) = \frac{10}{(s+1)(s+2)} = \frac{10}{s+1} - \frac{10}{s+2}$$

$$v_o(t) = 10(e^{-t} - e^{-2t})u(t)$$

Example. From poles and zeros to ODE: poles $-1 \pm j2$, zero -4, gain $K = 3$:

$$H(s) = \frac{3(s+4)}{s^2+2s+5} \quad \Rightarrow \quad y'' + 2y' + 5y = 3x' + 12x$$

System analysis

$$y(t) = h(t) * x(t) \qquad Y(s) = H(s)X(s)$$

Cascade: $H = H_1H_2$. Parallel: $H = H_3 + H_4$. Convert by factorising (cascade) or partial fractions (parallel).

Example.

$$H = \frac{4s+28}{s^2+6s+5} = \frac{4s+28}{s+1}\cdot\frac{1}{s+5} \text{ (cascade)} = \frac{6}{s+1} - \frac{2}{s+5} \text{ (parallel)}.$$

Poles, zeros and stability

Poles are the roots of the denominator of $H = P/Q$ (the characteristic equation $Q(s) = 0$, $H \to \infty$); **zeros** are the roots of P ($H = 0$). Plot poles as $\times$ and zeros as $\circ$. On the s-plane the horizontal axis is σ (damping) and the vertical axis is $j\omega$ (oscillation).

Pole location	Impulse response
Left half-plane, $H = 1/(s+a)$, $a > 0$	$e^{-at}u(t)$ decays: stable
Right half-plane, $H = 1/(s-a)$	$e^{at}u(t)$ grows: unstable
On the $j\omega$-axis	pure oscillation (no decay or growth), **not** BIBO stable
Origin ($1/s$)	constant (step at the output)
Real axis ($\omega_a = 0$)	no oscillation
Complex pair $\sigma_a \pm j\omega_a$	real part sets decay, imaginary part sets oscillation

For a complex pair the impulse response is

$$h(t) = \frac{1}{\omega_a}e^{\sigma_a t}\sin(\omega_a t)$$

BIBO stability (causal rational LTI) needs all poles strictly in the open left half-plane, $\mathrm{Re}\{p_i\} < 0$. Moving poles further left (same imaginary part) speeds the decay without changing the oscillation frequency; raising ω_a at fixed σ_a keeps the envelope and raises the frequency. A pole near the $j\omega$-axis gives a strong, slowly decaying response; a zero near it attenuates that frequency.

Example. Zeros and poles of

$$H(s) = \frac{(s+1)(s+3)}{[(s+1)^2+1](s+2)}$$

Zeros: $-1, -3$. Poles: -2 and $-1 \pm j$.
$H = (s^2+2s+2)/(s^2+4s+13)$: zeros $-1 \pm j$, poles $-2 \pm j3$.

Example. Find H from poles/zeros plus one value: poles $0, -2, -4$, zeros $-1, -3$, $H(1) = 4/3$: $H = K(s+1)(s+3)/(s(s+2)(s+4))$, $H(1) = K \cdot 8/15 = 4/3 \Rightarrow K = 2.5$.
Steps: factor N, factor D, find zeros, find poles, plot, interpret.

Example. $H = (s+2)/(s+5)$, unit step: $Y = \frac{1}{s}\frac{s+2}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5}$, so $y(t) = \frac{2}{5} + \frac{3}{5}e^{-5t}$. The pole at the origin gives the step; the pole at -5 gives e^{-5t}.

First- and second-order systems

First order: pole at $s = -a$, step input $X = 1/s$:

$$H(s) = \frac{a}{s+a} \qquad Y(s) = \frac{1}{s} - \frac{1}{s+a} \qquad y(t) = 1 - e^{-at}$$

The time constant is $\tau = 1/a$ (63% of the final value at $t = \tau$; initial slope $= a$), and a is the cutoff frequency. The further the pole from the $j\omega$-axis, the faster the transient.

Second order (standard form), with step response $Y = H/s$:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$\zeta > 1$ overdamped, $\zeta = 1$ critically damped, $0 < \zeta < 1$ underdamped with $\omega_d = \omega_n\sqrt{1-\zeta^2}$, $\zeta = 0$ undamped:

$$y(t) = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]$$

Lecture form $H = a/(s^2 + bs + a)$ (here $\omega_n^2 = a = 9$), step responses:

Case	b	Poles	$y(t)$
Overdamped	9	$-1.146, -7.854$	$1 + 0.171e^{-7.854t} - 1.171e^{-1.146t}$
Underdamped	2	$-1 \pm j\sqrt{8}$	$1 - e^{-t}\left(\cos\sqrt{8}t + \dfrac{\sqrt{8}}{8}\sin\sqrt{8}t\right)$
Undamped	0	$\pm j3$	$1 - \cos 3t$
Critically damped	6	-3 (repeated)	$1 - 3te^{-3t} - e^{-3t}$

Frequency response and filters

Steady-state response to $\sin(\omega t)$: replace s with $j\omega$. The gain is $|H(j\omega)|$ and the phase shift is $\Phi(\omega) = \angle H(j\omega)$.

$$y_{\mathrm{ss}}(t) = |H(j\omega)|\sin(\omega t + \Phi(\omega)) \qquad Y(\omega) = H(j\omega)X(\omega)$$

The cutoff ω_c is where $|H| = H_{\max}/\sqrt{2}$ (-3 dB). For the low-pass $H = a/(s+a)$:

$$|H(j\omega)| = \frac{a}{\sqrt{a^2+\omega^2}} \qquad \omega_c = a = \mathrm{BW}$$

$\omega = 0$: $|H| = 1$, $0°$; $\omega = a$: $1/\sqrt{2}$, $-45°$; $\omega \to \infty$: $|H| \to 0$, $-90°$.

Example. $H = 16/(s+4)^2$ (two cascaded poles): $|H(j\omega)| = 16/(\omega^2+16)$, $\Phi = -2\arctan(\omega/4)$. Near in phase at low ω; lag grows but saturates at $-180°$.

Circuit (output across)	$H(s)$	Phase $\Phi(\omega)$	ω_c
RC low-pass (C)	$\dfrac{1/(RC)}{s+1/(RC)}$	$-\arctan(\omega RC)$	$\dfrac{1}{RC}$
RL low-pass (R)	$\dfrac{R/L}{s+R/L}$	$-\arctan(\omega L/R)$	$\dfrac{R}{L}$
RC high-pass (R)	$\dfrac{s}{s+1/(RC)}$	$90^\circ-\arctan(\omega RC)$	$\dfrac{1}{RC}$

Low-pass: $|H(0)|=1$, passes below ω_c. High-pass $H=s/(s+\omega_c)$: $|H(0)|=0$ (blocks DC) and $|H(j\infty)|=1$, slope $+20$ dB/dec below ω_c.

Roll-off: each pole adds -20 dB/dec and each zero $+20$ dB/dec at high ω, so slope $=20(m-n)$ dB/dec (m zeros, n poles). 1st-order low-pass: -20 dB/dec; 2nd-order: -40 dB/dec (sharper). Cascading two identical first-order low-pass stages gives a 2nd-order low-pass with two repeated real poles.

Ideal blocks (Q: delay T, differentiator, integrator):

System	$H(s)$	$\lvert H(j\omega)\rvert$	Φ
Delay T s	e^{-sT}	1	$-\omega T$
Differentiator	s	ω	$+90^\circ$
Integrator	$1/s$	$1/\omega$	-90°

Second-order filters ($b_0=\omega_0/Q$; keep ω_0, raising Q moves poles along a circle of radius ω_0 towards the $j\omega$ -axis, sharpening the peak):

Type	$H(s)$	Key facts
Low-pass	$\dfrac{\omega_0^2}{s^2+b_0 s+\omega_0^2}$	$\lvert H(j\omega)\rvert=\omega_0^2/\sqrt{(\omega_0^2-\omega^2)^2+(\omega\omega_0/Q)^2}$; $\lvert H(j\omega_0)\rvert=Q$; -40 dB/dec. Real repeated poles: cutoff ω_0. Real distinct: cutoff set by the dominant pole (closest to $j\omega$-axis).
Band-pass	$b_0\dfrac{s}{s^2+b_0 s+\omega_0^2}$	zero at origin, complex poles; $\lvert H(j\omega_0)\rvert=1$; resonant-low-pass term ω_0^2 in numerator replaced by $b_0 s$
High-pass	$\dfrac{s^2}{s^2+bs+\omega_c^2}$	double zero at origin, $+40$ dB/dec; complex poles give a peak, real poles do not

Series RLC band-pass (output across R):

$$H(s)=\frac{(R/L)s}{s^2+(R/L)s+1/(LC)}$$

$$|H(j\omega)|=\frac{\omega R/L}{\sqrt{(1/(LC)-\omega^2)^2+(\omega R/L)^2}}\qquad \Phi=90^\circ-\arctan\left(\frac{\omega R/L}{1/(LC)-\omega^2}\right)$$

Cutoffs ω_l,ω_u are where $|H|=H_{\max}/\sqrt{2}$. The centre (resonant) frequency ω_0 is where H is purely real, $H_{\max}=|H(j\omega_0)|$ ($j\omega_0 L+1/(j\omega_0 C)=0$):

$$\omega_0=\frac{1}{\sqrt{LC}}=\sqrt{\omega_u\omega_l}\qquad \mathrm{BW}=\omega_u-\omega_l=\frac{R}{L}$$

$$Q=\frac{\omega_0}{\mathrm{BW}}=\frac{\omega_0 L}{R}\qquad |H(j\omega_0)|=1$$

$$\omega_{u,l}=\pm\frac{R}{2L}+\sqrt{\left(\frac{R}{2L}\right)^2+\frac{1}{LC}}$$

Q measures the passband width independent of its location.

Notch and band-stop (zero at the notch frequency ω_z, pole bandwidth $b_p=\omega_p/Q_p$; a notch is a band-stop with $\omega_p=\omega_z$, larger Q_p = narrower notch):

$$H(s)=\frac{s^2+\omega_z^2}{\omega_z^2}\qquad H(s)=\frac{s^2+\omega_z^2}{s^2+b_p s+\omega_p^2}$$

Pure notch: $|H(j\omega)|=|\omega_z^2-\omega^2|/\omega_z^2$, zero at ω_z (complex zero pair on the $j\omega$-axis).

Example. RC low-pass with $f_c=1$ kHz and $R=1.6$ kΩ:

$$f_c=\frac{1}{2\pi RC}\qquad C=\frac{1}{2\pi\cdot 1600\cdot 1000}=99.47\text{ nF}$$

Example. Band-pass with $R=10\Omega$, $L=1$ mH, $C=1\mu$F:

$$\omega_0=\frac{1}{\sqrt{LC}}=31.6\text{ krad/s}\qquad \mathrm{BW}=\frac{R}{L}=10\text{ krad/s}\qquad Q=3.16$$

$$\omega_u=37.0\text{ krad/s}\qquad \omega_l=27.0\text{ krad/s}\qquad \sqrt{\omega_u\omega_l}=\omega_0$$

Example. Notch for 50 Hz hum, $H=(s^2+\omega_0^2)/(s^2+2\omega_0\cos\theta s+\omega_0^2)$: $\omega_0=2\pi\cdot 50=314.2$ rad/s; $\theta=60^\circ$ gives $b_p=2\omega_0\cos 60^\circ=\omega_0$ ($Q_p=1$).

Op-amp (active) filters: inverting low-pass $H_1=-(R_f/R)\cdot a/(s+a)$, $a=1/(R_f C_f)$. With series input C: $H_2=-(C/C_f)(s+b)/(s+a)$, $b=1/(RC)$.

Pole-zero geometry and the magnitude / phase response

For $H(s)=K(s-z_1)\cdots(s-z_m)/[(s-p_1)\cdots(s-p_n)]$ put $s=j\omega$. Each factor $j\omega-r$ is a vector from the root r to the point $j\omega$ on the axis:

$$|H(j\omega)|=|K|\frac{\prod|j\omega-z_i|}{\prod|j\omega-p_k|}\qquad \angle H(j\omega)=\angle K+\sum\angle(j\omega-z_i)-\sum\angle(j\omega-p_k)$$

So $|H|$ is (product of distances to zeros) over (product of distances to poles), and the phase is (angles to zeros) minus (angles to poles). Slide a point up the $j\omega$ axis and watch the distances.

Feature	Effect on the response
Pole near the axis at ω_a	peak in $\lvert H\rvert$ near ω_a, sharper the closer it is; phase drops quickly there
Zero on the axis at ω_z	$\lvert H(j\omega_z)\rvert=0$ (notch); phase jumps by 180°
Zero near the axis	dip in $\lvert H\rvert$
Pole / zero at origin	$\lvert H(0)\rvert=\infty$ / $\lvert H(0)\rvert=0$
LHP real zero at $-a$ / pole at $-a$	phase $+\arctan(\omega/a)$ / $-\arctan(\omega/a)$, each moves 90° over $0\to\infty$
RHP zero (mirror of LHP zero)	same $\lvert H\rvert$, but phase lags instead of leads
Large ω	$\lvert H\rvert\sim\omega^{m-n}$: decays if $n>m$, grows if $m>n$

All-pass: a zero at $+a$ mirrored by a pole at $-a$ gives $|j\omega-a|=|j\omega+a|$, so $|H|=1$ for all ω and only the phase changes. **Quick roots:** $s^2+bs+c=(s+b/2)^2+(c-b^2/4)$, so $s=-b/2\pm\sqrt{b^2/4-c}$. For $s^2+2s+101$ this gives $-1\pm j10$.

Example. (a) $H=s/(s^2+2s+101)$: zero at 0, poles $-1\pm j10$. $|H(0)|=0$, sharp peak near $\omega=10$ where $|H(j10)|=10/\left(1\cdot\sqrt{401}\right)\approx 0.5$, then $|H|\approx 1/\omega\to 0$: band-pass. (b) $H=(s^2+25)/(s+1)$: zeros $\pm j5$, pole -1. $|H(0)|=25$, notch to 0 at $\omega=5$, then $|H|\approx\omega$ grows: notch plus high-pass. (c) $H=(s-2)/(s+2)$: all-pass, $|H|=1$.

Take-aways. Real part of a pole is decay or growth; imaginary part is oscillation frequency. Poles give the transient (natural) response; $H(j\omega)$ gives gain and phase against frequency. $X(\omega)$ is what is in the input, $H(j\omega)$ is how each frequency is treated, and $Y(\omega)=H(j\omega)X(\omega)$.